

Shoeb Ahmed Tanjim

Senior Software Architect

📍 Hamburg, Germany

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Summary

Passionate about Computer Programming, Distributed Systems, Software Architecture, and Computer Security with 6+ years of experience in end-to-end software development. A quick learner with experience in a wide range of tools & software stacks, and connecting the dots.

Skills

Languages: JavaScript, TypeScript, Python, C++, Java, PHP, Bash

Frameworks: FastAPI, Laravel, ExpressJS, Django, Flask, VueJS, NuxtJS, NextJS, Tailwind, ROS

Database & Message Broker: PostgreSQL, MySQL, SQLite, Firebase(NoSQL), LMDB, Redis, Kafka, RabbitMQ

Infrastructure & Cloud:

AWS: EC2, S3, Lambda, RDS, Aurora, EKS, ECR, IAM, SES, Cognito, Cloudwatch

Azure: VM, AKS, Managed DBMS, Blob, Backup, EntraID, Container Services, MSFoundry

Orchestration: Docker, Kubernetes (Both on-prem & cloud), Helm

IaC: CDK, Serverless, CloudFormation, Pulumi

Web: API, PWA, SPA, SSR, Nginx, Apache

Observability: Grafana, Prometheus, Loki, Promtail, Filebeat, Tempo, Otel Collector, Datadog, Sentry, Snyk

Others: Stackit, Bluehost, Cloudflare, Linux, Git, Github Action, Google SEO

Hardware & Robotics: ESP32, Raspberry Pi, Arduino, Jetson, Sensors

Project Management: Jira, Trello, ClickUp, Github Project

Work Experience

Senior Software Architect, Fehrmann MaterialsX GmbH

Jun 2025 - Present

- **Led the software development team**, translating high level business requirements into tasks and aligning with the development team.
- Designing the end-to-end architecture for SaaS based platform that offers cloud, on-prem, hybrid deployment depending on customers' needs.
- Developed and maintained several backend services and worker (ie: license manager, simulation runner)
- Designed and developed the entire cloud architecture in Azure for the product.
- TechStack: Kubernetes, LLM, RAG, Python, Azure, Pulumi, Terraform, Stackit

Senior Software Engineer, Provenant Inc, DE, USA

Oct 2022 - Jun 2025

- **Led the DevOps Guild**, responsible for implementing and enforcing DevOps best practices across projects, ensuring consistency, security & efficiency with company-wide operational goals.
- Received the '**Beyond Award**' for exceptional architectural design work on the Origin platform, addressing unique cybersecurity requirements and scaling constraints through independent initiative and effective implementation.
- Contributed from frontend written in NextJS to backend written in Java & Python.

- Developed a scalable CI/CD pipeline using GitHub Actions and Actions Runner Controller to build artifacts from multiple repositories, deploy to a Kubernetes cluster, and run unit and end-to-end tests, handling hundreds of jobs.
- Designed and implemented cost-effective, scalable & highly available Kubernetes cluster in Amazon EKS using Karpenter.
- Developed Helm charts and distributed them to customers via OCI-based registries.
- Implemented a cost-effective log & metrics collection & aggregation mechanism using Loki, Promtail, & Prometheus, collecting logs & metrics from all nodes & pods and storing them in EBS via PVC. Utilized node affinity to ensure Loki schedules in a specific availability zone.
- Implemented distributed tracing using Tempo with OpenTelemetry to collect tracing data from all services, integrating with Loki for log viewing and visualizing outputs in Grafana.
- Managing highly available stateful applications including Kafka cluster within Kubernetes environment.
- Developing microservices for cluster and configuration management which uses Dynamic Admission Control to inject configurations dynamically to services.
- Optimized Java memory consumption by over 40% in the development environment, enhancing developer experience (DX)
- Created custom email routing for the company and enforced email security through DKIM, DMARC, and SPF. Utilized advanced email monitoring tools to maintain domain reputation.
- Designed a secure network infrastructure spanning over 10 regions, utilizing peering connections to interconnect various networks.
- TechStack: MicroService Architecture, NextJS, Java, Python, AWS, Kubernetes, Helm, Docker, CI-CD, IaC(CDK), SSI, KERI, Bash, GithubAction

Associate Software Engineer, Dynamic Solution Innovators, Dhaka

Jun 2022 - Oct 2022

- Performed Reinforcement Learning algorithm analysis on various use cases, including path planning for autonomous vehicles, to optimize decision-making.
- Conducted R&D on autonomous systems, leveraging ROS, reinforcement learning (RL), and the Carla simulator to explore advanced robotics and self-driving technologies.
- Conducted R&D on autonomous vehicle software development using ROS & Gazebo simulator, implemented various path planning algorithms such as A*, and processed sensor data for real-time decision-making.
- Implemented 3D object detection using PointCloud data, leveraging various algorithms to improve accuracy and efficiency.
- Conducted sessions on various technical topics to enhance the skills and knowledge of team members across the company
- TechStack: ROS, ML(Supervised & Reinforcement Learning), C++, Python

Chief Technology Officer (CTO), Onushthan, Dhaka

Jan 2023 - July 2024

- Led the engineering team of Onushthan.com
- Analyzed business requirements and collaborated with the UI team to translate them into user interfaces, designing software that aligned with user needs and business objectives.
- Developed the backend using Express.js in TypeScript, integrated Prisma as the ORM, and utilized PostgreSQL as the database for efficient data management.
- Led project planning, distributed tasks among team members, and conducted code reviews to ensure high-quality code.
- Built cost-effective infrastructure in AWS to serve the application and integrated Datadog for collecting logs, and monitoring metrics.
- TechStack: NodeJS(ExpressJS), Typescript, Prisma, NextJS, AWS, Datadog

Full Stack Developer, BiTechX, Dhaka

Jan 2020 - Dec 2021

- Developed both the frontend using Vue and the backend using Laravel for the Hootstory application, including database design.
- Maintained the AWS infrastructure for the company, ensuring optimal performance, security, and cost-efficiency while supporting application needs.
- TechStack: VueJS, NuxtJS, NextJS/ReactJS, Laravel, AWS

Intern, Banglfire Solution Ltd, Dhaka

Feb 2021 - Mar 2021

- Developed an automated employee attendance system that collects information from WiFi-connected devices.
- TechStack: Python(nmap), Flask, Laravel

Education

Military Institute of Science and Technology, Dhaka, Bangladesh

Feb 2018 - Mar 2022

Degree: Bachelor of Science in Computer Science and Engineering

Projects

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- **Origin** *provenant.net*
Authenticates digital communications with the help of KERI. It uses a ledger-less approach for developing a self-sovereign identity system. This platform helps cryptographically sign and verify SMS & Voice calls and prove the origin.
TechStack: NextJS, Java, Python, Rust, MicroService, AWS, Kubernetes
 - **University Rover Challenge** *youtu.be/0ezyJEdleQM*
A rover that can traverse rough terrain, and establish communication with the base station over 1 km. The rover featured autonomous GPS-based navigation, a robotic arm for precise sample collection, and an integrated machine-learning system for real-time soil analysis.
TechStack: ROS, Supervised learning, Autonomous Navigation
 - **Drawbot** *youtu.be/DrkupUCWMu0*
A 3D mapping robot that can be operated via mobile/pc. It generates a 3D map of surrounding areas using a depth camera and sends the data back to the control station.
TechStack: Python, FFMPEG, ROS, Arduino, JavaScript, Socket Programming, Realtime Video Processing, PointCloud processing

To know more about my works: s-a-tanjim.netlify.app/works

Certifications

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- AWS Certified Solution Architect - Associate | Aug 2024
 - Python for Data Science | Jul 2021
 - CompTIA A+ | Jun 2018

Publications

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- Zaman, A., Majib, M. S., Tanjim, S. A., Siddique, S. M. A., Islam, Ashraf, F., ... & Rahman, M. M. "Phoenix: Towards Designing and Developing a Human Assistant Rover". IEEE Access, vol. 10, pp. 50728-50754, 2022, doi: 10.1109/ACCESS.2022.3170908.

- Zaman, A., Majib, M. S., Tanjim, S. A., Siddique, S. M. A., Islam, S., Aadeeb, M. S., ... & Islam, M. N. "UVC-PURGE: A Novel Cost-effective Disinfection Robot for combating COVID-19 Pandemic". IEEE Access, vol. 10, pp. 37613-37634, 2022, doi: 10.1109/ACCESS.2022.3163243.

Awards & Achievements

- **Champion** - University Rover Challenge 2021, Mars Society, USA
Team Co-Lead and Software & Communication Team Lead
- **Champion** - Hackathon For Nation | ICT Innovation Fest 2021, Dept of CSE, MIST
Developed Web-Based platform with PHP, JavaScript, HTML, CSS
- **Best Robotic System in Application Category** - Medical Robotics for Contagious Diseases Challenge 2020, Imperial College, London
TechStack of the Robot Software: ROS, Python, Socket Programming
- **MIST Silver Medal Winner | 2020**
For extraordinary performance in project development
- **2nd Runners Up** - Robowar | Digital Bangladesh Day 2018, A2i Innovation Lab
Developed software for the battle robot.
- **Best Junior Programmer** - Inter MIST Programming Contest 2018, MIST Computer Club
 - **Champion** - MIST Robo Fest 2018, MIST Robotics Club

Leadership & Extracurricular Activities

President - MIST Computer Club | 2021

- Served as President and Mentor of club, leading the club, organizing programs, and mentoring members through various classes and initiatives.

Co-Lead & Software and Communication Team Lead - MIST Mars Rover Society | 2021

- Developed the robot's software, leading the team to achieve top scores in the University Rover Challenge.

Vice President of Management & Instructor - MIST Robotics Club | 2021

- Organized many robotics competitions & classes to promote & encourage people in the field of robotics.
- Conducted classes on Robotics programming, working principles of different sensors, & circuit design.

Vice President & Instructor - MIST Computer Club | 2020

- Organized many nationwide programs like NCPC (National Collegiate Programming Contest), and IUPC (Inter-University Programming Contest).
- Conducted classes on different topics including fundamentals of programming, and web development.

Language Proficiency

- **Bangla:** Native Proficiency
- **English:** Full Professional Proficiency [IELTS Score: Band: 7]

Personal Interests

Astronomy, Physics, Psychology, Music, Travel, HomeLab